Last Name: ______________________
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UNIVERSITÉ DE GENÈVE
GENEVA SCHOOL OF ECONOMICS AND MANAGEMENT

Introduction to Macroeconomics

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Total number of pages: **9 pages + 1 answer sheet**

Duration of the exam: 120 minutes (30 questions)

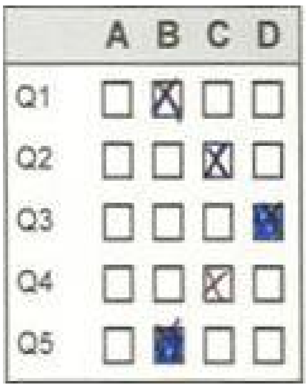
Authorized material: none (calculator is not authorized).

Instructions to answer multiple choice questions:

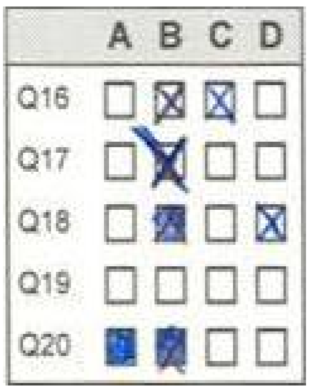
- ⋆ The answer sheet will be distributed when we proceed to the identity check. Please leave your student card on the table in front of you to facilitate the identity check.
- ⋆ On this page **AND** in the answer sheet, write down clearly and in capital letters your family name, first name and student number (= 8 digit number without dashes)
- ⋆ The questionnaire **AND** the answer sheet of the exam must absolutely be handed in at the end of the exam.
- ⋆ On the answer sheet, the box corresponding to the chosen answer (A to D) must be checked or filled properly with a dark blue or black pen.
- ⋆ If necessary, make corrections carefully. The answer must be clearly identifiable and unique. If this is not the case, it will be counted as wrong.
- ⋆ We do not answer questions during the exam.
- ⋆ WC exits are **unauthorized** during the exam.

Example:

Good



Bad



In order not to waste your chances, answer all questions.
There are no negative points and there is only one correct answer.

Question 1

In macroeconomics, the savings paradox refers to the fact that :

(a) foreign savings can fund domestic investment.

(b) even it is rational at the individual level to increase savings, this behavior can hurt the economy at the aggregate level.

(c) any investment is necessarily the outcome of savings of another individual creditor, because for every debtor there is a creditor.

(d) when the private sector does not save, the government will need to save instead for investment to be positive.

Question 2

To move from GDP to GNP we need to take into account :

(a) production subsidies.

(b) capital depreciation.

(c) capital income.

(d) net factor income abroad.

Question 3

Nominal GDP decreased from 220 billion to 210 billion, while real GDP decreased from 220 billion to 200 billion. As a consequence :

I. prices have increased.

II. prices have decreased.

III. quantities have increased.

IV. quantities have decreased.

Which of the following is correct ?

(a) Propositions I and III are correct.

(b) Propositions I and IV are correct.

(c) Propositions II and III are correct.

(d) Propositions II and IV are correct.

Question 4

Which of the following statements regarding the measurement of well-being is correct ?

(a) GDP per capita is an objective and ideal indicator of well-being.

(b) GDP per capita is positively correlated with variables capturing the quality of life.

(c) GDP per capita is a measure of income inequality.

(d) The *Human Development Index* (Index of Human development project) does not take into account GDP per capita.

Question 5

Consider a world with **only two countries** that trade with each other :

Country A		Country B	
Consumption	100	Depreciation	30
Gross investment	40	Subsidies	20
Government Expenditure	50	Indirect Tax	10
Exports	60	National Income	160
Imports	30		
Factor income paid abroad	20		
Factor income from abroad	30		

In addition, we know that in Country A, depreciation is equal to subsidies minus indirect taxes. Which of the following statements is correct ?

(a) NNP at factor prices of Country A is 230 and GDP at market prices of Country B is 190

(b) National Income of Country A is 220 and GDP at factor prices of Country B is 200

(c) National Income of Country A is 230 and GDP at market prices of Country B is 200

(d) GNP at market prices of Country A is 220 and GDP at market prices of Country B is 190

Question 6

Which of the following statements concerning the consumer price index (CPI) and the GDP deflator is correct ?

(a) The CPI adjusts the composition of a basket of goods and services to reflect changes in the production structure while the GDP deflator focuses on a basket of goods produced in the economy.

(b) The GDP deflator focuses on goods produced within the economy while the CPI focuses on a typical consumption basket.

(c) Since the GDP deflator does not take into account the price of imported goods, it is always smaller than the CPI.

(d) There is no difference between the CPI and the GDP deflator : they both measure the rate of inflation.

Question 7

The CPI is falling. Which of the following is correct ?

(a) The economy is experiencing stagflation.

(b) Indexed contracts such as wages and pensions will increase.

(c) For the central bank, this is an argument in favor of tighter monetary policy.

(d) With constant nominal interest rates, incentives to save are increasing.

Question 8

The empirical evidence suggests that we observe convergence of GDP per capita between countries :

(a) when capital in those countries is scarce.
(b) when political institutions and other socio-economic characteristics of these countries are quite similar.
(c) when countries are rich in natural resources.
(d) when countries are rich in human capital.

Question 9

Which of the following is correct ?

(a) In the presence of constant returns to scale, an increase in the number of workers due to immigration will increase GDP per capita.
(b) If factors of production exhibit decreasing marginal productivity, doubling the amount of all factors of production can double production.
(c) In the presence of increasing returns to scale, an increase in the number of workers due to immigration will increase GDP per capita.
(d) If all factors of production exhibit decreasing marginal productivity, then there are decreasing returns to scale.

Question 10

The unemployment rate increases from 9% to 10%. What can we conclude with certainty ?

(a) The number of persons searching for a job has increased.
(b) The labor force has decreased.
(c) If the labor force has increased, the number of persons searching for a job also increased.
(d) None of the above.

Question 11

Frictional unemployment **depends on** :

(a) the setting of efficiency wages by companies.
(b) the introduction of a minimum wage at the national level.
(c) seasonal changes in labor demand from businesses.
(d) Search and matching costs.

Question 12

Which of the following factors could explain a decline in the demand for loanable funds ?

(a) A reduction in dividend taxes.
(b) A reduction of a tax credit for upgrading equipment.
(c) A reduction of the interest rate.
(d) A reduction of a subsidy to improve the insulation of buildings.

Question 13

Which of the following is correct ?

(a) M_0 includes coins, banknotes, and bank accounts and deposits.

(b) When the Central Bank buys bonds, it reduces money supply.

(c) When the Central Bank reduces the interest rate, money supply increases.

(d) Everything else equal, if economic agents decide to keep a larger share of their savings outside the financial system, the money supply increases.

Question 14

The Swiss National Bank buys Euros for 10,000 francs. Knowing that the reserve requirement ratio is 10 % and households do not hold cash (notes and coins), what is the variation in the Swiss money supply M1 ?

(a) -100'000

(b) 10'000

(c) -10'000

(d) 100'000

Question 15

Which of the following is correct ?

(a) A currency's purchasing power determines the amount of money in circulation.

(b) Most economists claim that the money supply has a long-term impact on the so-called real economic variables.

(c) According to monetarists, the amount of money circulating in the economy has no long-term impact on investment decisions.

(d) The lack of correlation between inflation and growth in money supply illustrates the concept of long-run money neutrality.

Question 16

The economy of a country can be described by the following statistics : nominal transactions are 500, the price level is 10 and the velocity of money is 5. According to the quantitative equation of money, the money supply is :

(a) 50

(b) 100

(c) 500

(d) 1000

Question 17

If the money supply, the velocity of money and real GDP grow at the same rate, inflation is :

(a) Zero.

(b) Positive.

(c) Negative.

(d) None of the above.

Question 18

In an open economy, which of the following is correct ?

(a) An improvement in the trade balance implies a smaller net capital outflow.

(b) If the balance of payments is in equilibrium, a positive net factor income abroad necessarily finds its counterpart in the trade balance.

(c) All things being equal and assuming that net factor income abroad is zero, an imbalance in the trade balance necessarily finds its counterpart in the capital account balance.

(d) If the trade balance is fixed, a decrease in net factor income abroad implies an increase in net capital outflows.

Question 19

In an open economy with a large surplus in the trade balance, which of the following is necessarily true ?

(a) Investment is greater than national savings.

(b) Net capital outflow is negative.

(c) Public saving is positive (i.e., there is a government surplus).

(d) Investment is smaller than national savings.

Question 20

The loss of foreign-investor confidence in a country's economy causes severe capital flight. At the new long-run equilibrium, which of the following is correct ?

(a) The national currency will appreciate, the interest rate and net exports will increase.

(b) The national currency will depreciate, the interest rate and net exports will increase.

(c) The national currency will appreciate, the interest rate will rise and exports will decrease.

(d) The national currency will depreciate, the interest rate and net exports will decrease.

Question 21

According to open macroeconomics model seen in class, if a government returns to a balanced budget after being in deficit, you can expect that :

(a) The supply of loanable funds increases, decreasing the interest rate and increasing the net capital outflow, which implies a depreciation of the national currency and increase in net exports.

(b) The demand for loanable funds increases, increasing the interest rate and decreasing the net capital outflow, which implies an appreciation of the national currency and an increase in net exports.

(c) The supply of loanable funds increases, decreasing the interest rate and increasing the net capital outflow, which implies an appreciation of the national currency and a decrease in net exports.

(d) The demand for loanable funds decreases, decreasing the interest rate and increasing the net capital outflow, implying a depreciation of the national currency and a decrease in net exports.

Question 22

Let an open economy be characterized by the following :

Net capital outflows	-10
Net factor income abroad	0
Government expenditure	100
Taxes	50
Consumption	50
Private Savings	60

Which of the following is correct ?

(a) Investment is 0

(b) Investment is 10

(c) Investment is 20

(d) Investment is 30

Question 23

According to the relative PPP theory for the determination of exchange rates, if price levels in China increase at an annual rate of 8% , and decrease at an annual rate of 2% in Switzerland, we can conclude that there will be :

(a) a 6% appreciation of the CHF with respect to the Yuan.

(b) a 6% depreciation of the CHF with respect to the Yuan.

(c) a 10% appreciation of the CHF with respect to the Yuan.

(d) a 10% depreciation of the CHF with respect to the Yuan.

Question 24

Consider a closed economy where the marginal propensity to save is 10 %, income is 100 and the government budget is balanced. The government decides to increase spending by 10 while keeping its budget balanced (so it increases its lump sum tax also by 10). According to the Keynesian multiplier theory (and in the absence of crowding out), what will happen ?

(a) Consumption increases by 9

(b) Income increases by 9.

(c) Consumption increases by 100.

(d) Income increase by 100.

None of the answers are correct here. Consumption will remain unchanged because disposable income is unchanged. And income increases by 10 because the keynesian multiplier of the balanced budget is equal to 1, so the change in income equals the change in government expenditure (which itself equals the change in goverment tax revenue)

Question 25

Which of the following **is not** compatible with a positive relationship between the general level of prices and GDP in the short run (short-run aggregate supply) ?

(a) The existence of rigid wages that maintain high costs for firms despite a general decline in the price level.

(b) A short run decrease in employment in the economy as a result of a disinflation policy of the central bank.

(c) A restrictive monetary policy intervention perfectly anticipated by economic agents.

(d) A misperception of firms who believe that the decline in overall prices represents a decrease of the specific demand for their products.

Question 26

In a liquidity trap :

(a) an increase in government expenditure is more likely to succeed to revive the economy than a reduction of the interest rate by the Central Bank.

(b) the interest rate is too low.

(c) price adjust in the short-run and the Central Bank cannot influence the economy's real variables by increasing money supply.

(d) individuals prefer to hold financial assets such as bonds rather than cash.

Question 27

According to the aggregate demand/aggregate supply model, we can conclude with certainty that aggregate demand will shift to the right when :

(a) government expenditure and taxes increase.

(b) the marginal propensity to consume increases.

(c) prices increase.

(d) prices decline.

Question 28

Which of the following causes a shift of the Phillips curve in the short term ?

(a) An increase in public spending.

(b) An increase in private consumption due to a wave of optimism related to the Football World Cup.

(c) A destruction of communication channels due to an earthquake.

(d) A decrease in the money supply by the central bank that does not affect inflation expectations.

Question 29

A disinflation policy of the central bank :

(a) does not have a transitory effect on the real economy because of the neutrality of money.

(b) helps stimulate the economy in a period of economic crisis.

(c) implies a temporary reduction in unemployment in the economy.

(d) leads to a shorter decline in GDP if the central bank is credible.

Question 30

In the absence of international cooperation during an economic crisis :

(a) we observe an underprovision of fiscal stimulus, because each country hopes to take advantage of the efforts of other countries without bearing the costs.

(b) aggregate demand does not react to an increase in public expenditure and fiscal stimulus loses its effectiveness.

(c) a protectionist economic policy allows for an optimal allocation of resources to the most efficient firms that will create more jobs.

(d) the costs needed to stimulate the economy are the same as in the case of international coordination.